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Ensemble of Markovian stochastic dynamic programming models in different time scales for long term hydropower scheduling

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ABSTRACT

This paper presents a new approach for long term hydropower scheduling. In opposition to the standard Markovian stochastic dynamic programming approach, where monthly inflows are modeled according to probability distribution functions, conditioned to some occurrence of inflow in the previous month, in the proposed approach the monthly inflows are aggregated in different time scales and then submitted to the Markovian model. The discharge decisions are then calculated by a deterministic model that optimizes the problem for one year ahead according to inflows provided by a combination of each Markovian model. Tests were conducted on hypothetical single-reservoirs hydrothermal systems using data from four real Brazilian hydro plants, with distinct hydrological regimes. The performance of the proposed method was evaluated through simulation, using the historical inflow data, in comparison with the standard Markovian model. The results have shown that the proposed approach has provided spillage reduction and increase on hydro productivity as well as power generation, which incurred in up to 2.1% reduction in operational costs.

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1. Introduction

The idea of controlling reservoir's storage is connected to the human development since ancient times [1]. After 1940's, accelerated by the second war, a large advance on dam management was published emerging from the theory of sequential decisions [2].

Later on, many contributions were proposed and the most accepted technique, until nowadays, to solve long term hydropower scheduling, LTHS, was proposed in 1976. Stochastic dynamic programming, SDP, is a closed-loop control policy that aims to determine optimal decisions for a range of discrete states of the system. The main limitation of SDP is the so-called "curse of dimensionality" as the computational effort increases exponentially with the number of state variables [3].

As scheduling problems were getting more challenging (power system growth, diversification of power sources and market issues involved) and taking advantage of technological advances, several improvements were developed over the SDP theory.

In Ref. [4] general uses of dynamic programming, DP, and classical modeling are presented. Later on, some adaptations to bypass the "curse of dimensionality" [12] and to solve instability problems [13–15] were implemented. Different approaches and heuristics methods, associated with DP [7,10] were also considered for different problems. Intelligent systems with neural networks were also incorporated to DP in Refs. [5,6,8,9,11]. A deep resume of a great number of DP based techniques for solving reservoirs operation problems is presented in Ref. [16]. Streamflow prediction models were also tested in substitution of classical statistical models [17].

Annual inflow prediction has been proposed and tested in a framework of predictive control modeling [18]. The comparison between this method and the standard one, Markovian stochastic dynamic programming, MSDP, has shown expressive improvements, indicating that annual inflow models can provide better information for optimal monthly decision-making problems. The reason is that it might be more important to foresee the total inflow for the next year, which includes the next wet season, then to foresee the inflow of the next month [20–24], since optimal decision-making aims to maximize hydro generation but at the same time distribute it on time so that thermal generation costs be as flat as possible within the planning horizon. The extension of this idea to SDP is possible and has shown promising results [19].

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This paper proposes a new approach to solve LTHS problems based on a nested composition of MSDP models which results, from the aggregation of monthly inflows up to one year on different time scales. Decision-making at each month is then determined as the optimal decision of the first monthly stage of a deterministic nonlinear optimization model that solves the LTHS one year ahead, in monthly steps, with inflows provided by the MSDP models. It is possible to implement the proposed technique in a larger system with more than one hydropower plant using the equivalent reservoir technique [4]. In this paper only single reservoir systems were considered to clearly identify the differences between the classical methods and the proposed one.

This paper is structured as follows. Section 2 presents the LTHS model with a brief explanation of the SDP modeling and main variables description. The proposed method is also explained and discussed, along with the motivation and some characteristics of the hydro data that support the main idea. Section 3 presents the case studies and results. Finally, Section 4 summarizes the main contribution and takes some conclusions.

2. Problem formulation

The LTHS for single-reservoir hydropower systems can be formulated as the following stochastic optimization problem:

$$\text{Objective: } \alpha_1(x_0, y_0) = \min_q E_y \left\{ \sum_{t=1}^{T} \psi(d_t - g_t) \right\} \quad (1)$$
$$0$$

$$\text{Subject: } g_t = k.h_t.q_t \quad (2)$$

$$h_t = \phi\left(x_t^{avg}\right) - \theta(u_t) - \xi \quad (3)$$

$$x_t^{avg} = (x_{t-1} + x_t)/2 \quad (4)$$

$$u_t = q_t + s_t \quad (5)$$

$$x_t = x_{t-1} + (y_t - u_t).\delta_t \quad (6)$$

$$X^{min} \le x_t \le X^{max} \quad (7)$$

$$q_t \le Q^{max} \quad (8)$$

$$u_t, q_t, v_t \ge 0 \quad (9)$$

where E_y, is the expected value with respect to inflows; t, time stage index (months); T, number of time stages in planning period; ψ, thermal generation cost ($); d, load demand (MW); g, hydropower generation (MW); k, constant efficiency factor (MW/{m^3/s m}); h, net water head *(m)*; x, reservoir storage (hm^3); u, release from reservoir (m^3/s); q, discharge through turbines (m^3/s); s, spillage from reservoir (m^3/s); $X^{min,max}$, limits for reservoir storage (hm^3); Q^{max}, maximum discharge (m^3/s); ϕ, forebay elevation function (m); θ, tailrace elevation function (m); ξ, average penstock head loss (m); y, inflow into reservoir (m^3/s); δ, stage duration (s/10^6).

The objective function, α, (1) aims to minimize over a planning horizon T, the expected value of ψ, expressed as a quadratic function, mainly related with the fuel used at the thermal plants. As d is fixed, these costs, are controlled by g, which is a product of k, h, and q. The constant k includes the water density, gravity acceleration and average turbine/generator efficiency. Two different polynomial functions, up to fourth degree, are used to calculate ϕ and θ.

The variables u, q and s are non-negative and defined for every t. The maximum limit was defined only for q since s is not handled as a decision variable during optimization. The water balance, or state transition, Eq. (6) determines x_t using the previous (or initial) reservoir storage, $x_{(t-1)}$, y, (inflow) and u, (outflow). The conversion constant δ only converts cubic meter per second in cubic hectometer per month. Terms such as evaporation and other water uses have not been considered for the sake of simplicity.

2.1. Markovian stochastic dynamic programming

MSDP determines rules for decision making at each stage of the planning period which will provide the optimal decision for each possible state of the system. Mathematically, the MSDP technique determines a sequence of decision functions, $q^*_t(x_{t-1}, y_{t-1})$, mapping the possible states (reservoir storage, past inflow), onto decisions (discharge) that minimize the expected costs.

The MSDP considered here assumes that the stochastic variable, y_t, depends only on the inflow of the previous stage, y_{t-1}. This means that inflows are represented by a first-order periodic autoregressive model, PAR-1, what makes it possible to describe such uncertainty as a Markov chain process.

For each stage, decisions are ranked based on the minimization of the expected sum of the present cost and the future cost, assuming optimal decision-making for all subsequent stages, according to Bellman's Optimality Principle. Therefore the problem can be solved in a recursive way:

$$\alpha_t(x_{t-1}, y_{t-1}) = \min_q E_y \left\{ \psi(d_t - g_t) + \alpha_{t+1}(x_t, y_t) \right\} \quad (10)$$

As we are interested on the control law of the system on stead state, the backward recursion can be computed, starting from the last stage where terminal cost is assumed null, and must be performed until the convergence of the monthly decision functions. This procedure requires, at least, the discretization of the state variable. The control variable can either be discretized or treated as a continuous variable.

Normal (Gaussian) and Log-normal distribution are the most widely used probability density functions (PDF) for inflows modeling [25]. Let y_m represent the inflow data for every month, m = [1,2,...12], the log-normal conditional density distribution function of the inflows, estimated from historical records, can be expressed as in Eq. (11), where ρ_m represents the correlation between two consecutive months, $\mu_{(m|m-1)}$ the conditional mean (12) and $\sigma_{(m|m-1)}$ the conditional standard deviation (13).

$$f(y_m|y_{m-1}) = \frac{1}{\sqrt{2\pi\sigma^2_{m|m-1}}} e^{-\left(\frac{\hat{y}_t - \mu_{m|m-1}}{\sqrt{2}\sigma_{m|m-1}}\right)^2} \quad (11)$$

$$\mu_{m|m-1} = \mu_m + \rho_m \frac{\sigma_m}{\sigma_{m-1}} \left(\hat{y}_{t-1} - \mu_{m-1}\right) \quad (12)$$

$$\sigma_{m|m-1} = \sigma_m \sqrt{1 - \rho_m^2} \quad (13)$$

Conditional probability of discrete inflow ranges, $y_t \in$ (a,b], represented by their average value, y_t^i, can be calculated as in Eq. (14):

$$P\left(y^i_{t|t-1}\right) = \int_a^b f(y_t|y_{t-1})\, dy \quad (14)$$

Finally, the expected cost can be calculated, according to the conditional probabilities of inflows being in the interval represented by y_t^i from N possible ranges, as expressed below:

$$E_{y_{t|t-1}} \left\{ \psi(d_t - g_t) + \alpha_{t+1}(x_t, y_t) \right\}$$
$$= \sum_i^N \left\{ \psi(d_t - g_t) + \alpha_{t+1}\left(x_t, y_t^i\right) \right\} P\left(y^i_{t|t-1}\right) \quad (15)$$

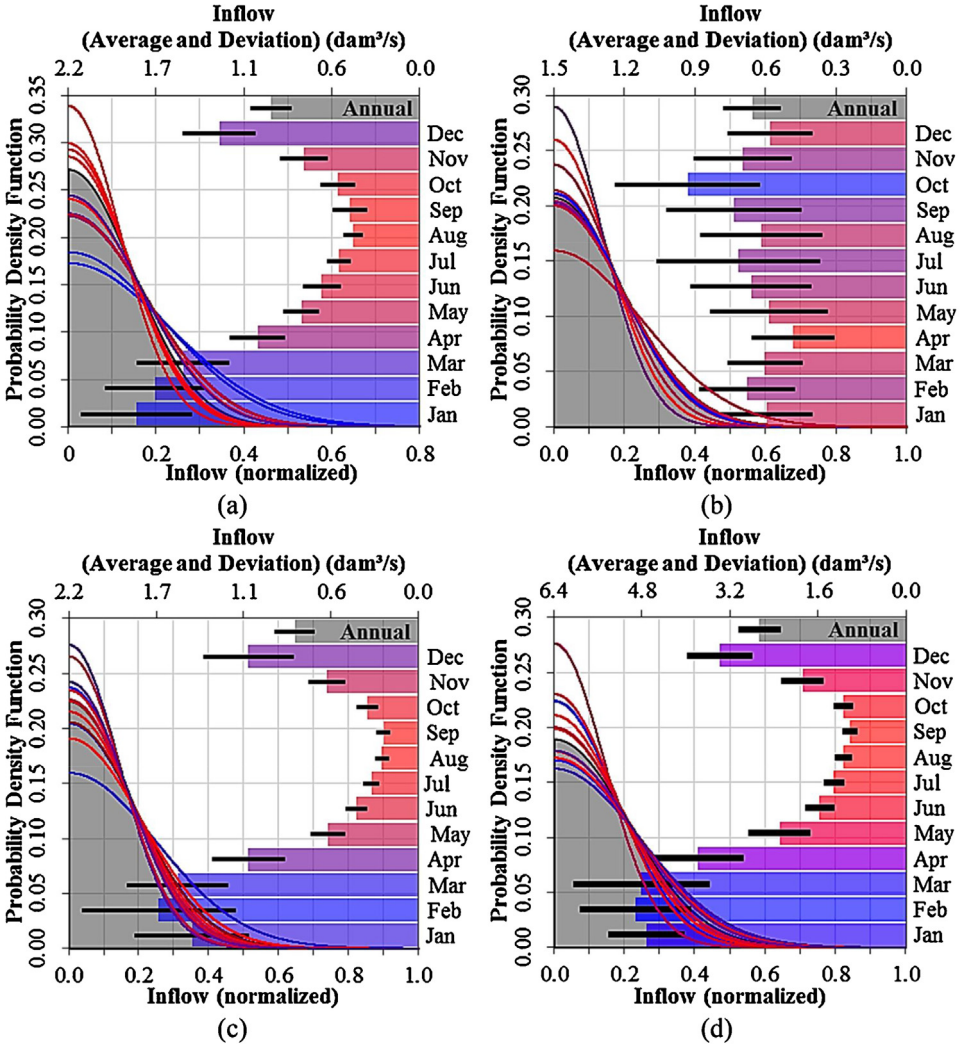


Fig. 1. Normal probability density function, average and standard deviation (monthly and annual) for: (a) Furnas, (b) Foz do Areia, (c) Serra da Mesa and (d) Sobradinho.(For interpretation of the references to color in thefigure text, the reader is referred to the web version of this article.)

2.2. Annual aggregated inflows

Four hydro plants, HP, were chosen to compare the single reservoir case studies, each of them from different Brazilian river basins. Fig. 1 presents data that support the main idea of this paper. The curves on the left hand side are the PDF in normal distributions, where normalized average values, cross the vertical axis and the negative values were suppressed. The PDF for annual data was emphasized with its area painted gray.

The colored bars, on the right hand side, represent the monthly average inflows, along with the annual average inflow, in gray. The horizontal black bars represent the standard deviation values. The color shades from blue to red is proportional to the average inflow: the darker blue one represents the wettest month and the darker red one represents the driest month.

This color gradation aims to illustrate the difference between seasons with smooth or rough transitions depending on the river. Hydrological regimes shown in Fig. 1(a), (c) and (d) are very similar, with well-defined wet and dry periods and higher standard deviation on wet months. Meanwhile the one presented in Fig. 1(b) is very distinct. The difference between the wettest month, October, and the driest month, April, is quite expressive. Besides, the standard deviations are greater when compared to the respective average inflows, indicating that this river has a much more variable hydrological regime.

Comparing these monthly data with the annual one, we can observe that the annual standard deviation is smaller than the majority of monthly standard deviations (exceptions occur during the dry season). Statistically we can calculate the coefficient of variation, which is the quotient of standard deviation and average, shown in Fig. 2.

Comparing the absolute values of standard deviation and average inflows in Fig. 1, it is clear that annual values are lower than those of wet months but higher than those of dry months. Fig. 2 shows the coefficient of variation that summarizes, in a single measure, standard deviation and average of inflows. With these data it is possible to analyze several possible aggregations and observe that this coefficient decreases as more months are aggregated. This means that an annual MSDP model could perform relatively better than a monthly one since the uncertainty of inflows on annual basis is relatively smaller than that on monthly basis.

Indeed, improvements on MSDP performance applied to LTHS can be expected for any time scale of inflow aggregation, consequently reducing the standard deviation as the time scale resolution increases.

Based on this assumption, a MSDP^b model is implemented for a certain number, b, of months aggregated with $b \in \{1,2,3,4,6,12\}$,

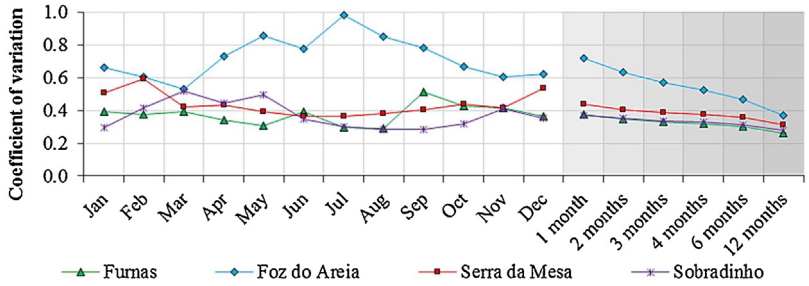


Fig. 2. Coefficient of variation for month inflows and average of 1, 2, 3, 4, 6 and 12 months aggregated.

Table 1
Hydro plants operational bounds.

Hydro plants	P^{max} (MW)	ΔX (hm^3)	Q^{max} (m^3/s)	k (MW/m^3/s m)
Furnas	1312.0	17217.0	1402.7	0.008633
Foz do Areia	1676.0	3805.0	1222.4	0.009025
Serra da Mesa	1275.0	43250.0	1094.8	0.009134
Sobradinho	1050.0	28669.0	3600.8	0.009023

so that corresponding decision rules, $q_t^b(x_{t-1}, y_{t-1})$, can be obtained. Note that $q_t^1(x_{t-1}, y_{t-1}) = q_t^*(x_{t-1}, y_{t-1})$ is the decision rule obtained by traditional MSDP[1] on monthly basis. Associated with each MSDP^b model there is a lag-one periodic autoregressive model PAR^b(1) that provides the conditioned probability of inflow, y_t^b, in the corresponding aggregation.

The MSDP-A (MSDP Aggregated) model here proposed is based on a nested composition of MSDP^b models. The decision making on MSDP-A at each stage, t, is obtained by a deterministic optimization model that solves the LTHS for a year ahead horizon with parameters that take into account the decisions q_t^b provided by the MSDP^b model in time scale b according to some expectation of inflows $\hat{y}_t^b$ provided by the PAR^b(1) model.

The first parameter necessary for the annual deterministic optimization is the inflow forecasting $\hat{y}^j$, $\{j \in Z/1 \le j \le 12\}$, for the next twelve months ahead. This is provided by a nested composition of the inflow forecast in different time scale resolutions, as follows:

$$\hat{y}_t^{t+1} = \hat{y}_t^1 \tag{16}$$

$$\hat{y}_t^{t+2} = \hat{y}_t^2 \left(\frac{\bar{y}^2}{\sum_{l=1}^{2} \bar{y}^l} \right) \tag{17}$$

$$\hat{y}_t^{t+3} = \hat{y}_t^3 \left(\frac{\bar{y}^3}{\sum_{l=1}^{3} \bar{y}^l} \right) \tag{18}$$

$$\hat{y}_t^{t+4} = \hat{y}_t^4 \left(\frac{\bar{y}^4}{\sum_{l=1}^{4} \bar{y}^l} \right) \tag{19}$$

where $\bar{y}^j$ is the monthly average inflow, j months ahead of t. The inflow forecasting for the next two months, $\hat{y}_t^{t+j}$, $j = \{5, 6\}$ can be calculated by:

$$\hat{y}_t^{t+j} = \hat{y}_t^6 \left(\frac{\bar{y}^j}{\sum_{l=1}^{6} \bar{y}^l} \right) \tag{20}$$

The inflow forecast for more than 6 months ahead, $\hat{y}_t^{t+j}$, $j = \{7, \ldots, 12\}$ are calculated by:

$$\hat{y}_t^{t+j} = \hat{y}_t^{12} \left(\frac{\bar{y}^j}{\sum_{l=1}^{12} \bar{y}^l} \right) \tag{21}$$

The other parameter necessary for the annual deterministic optimization is the final storage of the reservoir at the end of the year ahead, which is calculated by water balance:

$$x_{t+12} = x_t + \sum_{l=1}^{12} \hat{y}_t^{t+l} * \delta_{t+l} - q_t^1 * \delta_{t+1} - q_t^2 * \delta_{t+2} - q_t^3 * \delta_{t+3} - q_t^4 * \delta_{t+4} - q_t^6 * \sum_{j=5}^{6} \delta_{t+j} - q_t^{12} * \sum_{j=7}^{12} \delta_{t+j} \tag{22}$$

A sequential quadratic programming (SQP) [26] algorithm provides the solution of the deterministic model from which the discharge decision of the first month is taken as the optimal decision of the proposed approach. Different combinations of the optimal decisions and future inflows forecast for the MSDP models adjusted for different time scales could be proposed. The central idea is to ensemble different views of the future inflows and water discharges according to MSDP models at different time scales.

3. Case study

The simulation case studies were conducted for single reservoir hydrothermal systems involving four Brazilian hydro plants with distinct hydrological regimes. The hydro plant's data is available online[1] and is summarized in Table 1, where ΔX is the active reservoir storage capacity and P^{max} is the installed capacity of the hydro plant.

The load demand considered in all simulated hydrothermal systems is constant and equal to the installed capacity of the corresponding hydro plant. Fig. 3 shows the storage levels for Furnas (a), Foz do Areia (b), Serra da Mesa (c) and Sobradinho (d) hydro plants for the planning period from May, 1932 to April, 2012, grouped by month, along with the corresponding average. The perfect foresight solution, PF, is also presented as it represents an upper bound for the results.

The continuous lines, in Fig. 3, are the average values per month. Higher storage values are interesting because they increase the water head and, consequently, the plants efficiency providing, therefore, greater power generation with the same discharge, although, in simulation time (water balance), they can incur in water spillage which is something that optimal operation, indi-

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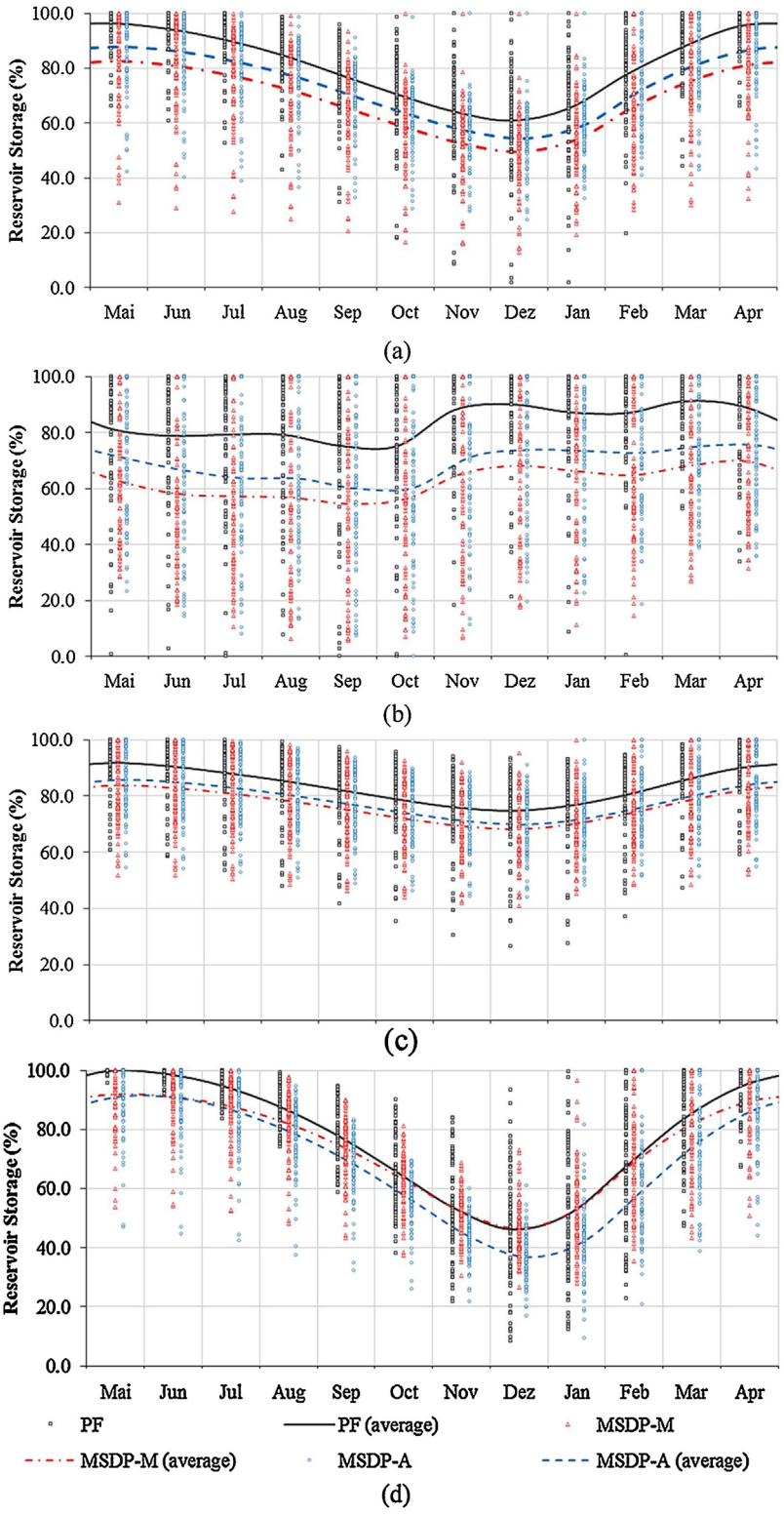


Fig. 3. Reservoir storage (planning period and average) per month for hydro plants.

Table 2
Simulation results – water releases and efficiency for Furnas HP.

	Discharge (m^3/s)	Spillage (m^3/s)	Efficiency (%)	Productivity ($MW/m^3/s$)
DP – PF	918.2	5.6	99.39	1.0691
MSDP – M	896.4	28.4	97.03	1.0212
[%]	−2.37	407.3		−4.48
MSDP – A	898.8	25.8	97.30	1.0306
[%]	−2.11	360.8		−3.60

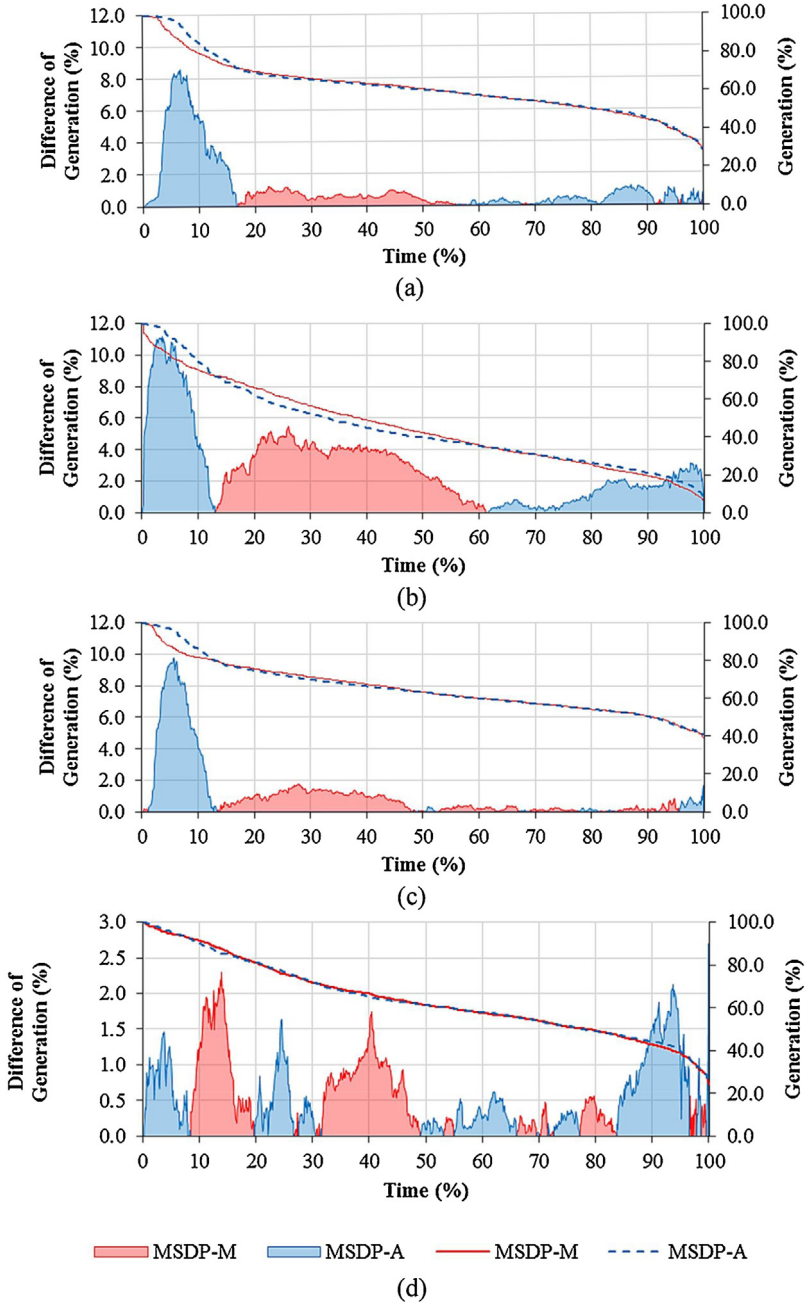


Fig. 4. Power generation permanency curves and differences between MSDP-M and MSDP-A.

rectly, try to avoid. The permanency curves of hydro generation, shown in Fig. 4, are organized in descending percentage.

The numerical differences between MSDP-M and MSDP-A permanency curves are difficult to visualize, and for that reason Fig. 4 also shows the positive percentage difference between the techniques, (highest values minus lowest).

Different hydro plants have presented different generation patterns as can be observed in Fig. 4. Although for all case studies it can be observed that a better performance was provided by the MDSP-A at the higher generation zone and also some improvements at the lower generation zone can be observed as well.

For Furnas (a) and Serra da Mesa (c) hydro plants more expressive improvements were provided by the MSDP-A at the higher generation zone, with quite small differences at mid and low generations. In Foz do Areia (b) power plant, specially, the differences are up to 10% at the higher generation zone, but significant variations were also observed at mid and low generation zones. For Sobradinho hydro plant (d) both approaches presented quite similar result, which explains the poor improvement of MSDP-A.

Table 3
Simulation results – operation cost and hydro generation for Furnas HP.

	Cost ($)	Hydro generation (MW)	
		Average	Standard deviation
DP – PF	3.71	726.9	143.5
MSDP – M	4.09	701.0	162.9
[%]	10.18	−3.57	13.56
MSDP – A	4.04	708.4	174.0
[%]	8.73	−2.54	21.29

Table 4
Simulation results – water release and efficiency for Foz do Areia HP.

	Discharge (m^3/s)	Spillage (m^3/s)	Efficiency (%)	Productivity ($MW/m^3/s$)
DP – PF	637.7	16.2	97.52	1.4482
MSDP – M	573.1	81.5	87.64	1.3137
[%]	−10.13	401.7		−9.29
MSDP – A	580.7	73.5	88.80	1.3605
[%]	−8.94	352.7		−6.05

Table 5
Simulation results – operation cost and hydro generation for Foz do Areia HP.

	Cost ($)	Hydro generation (MW)	
		Average	Standard deviation
DP – PF	9.73	751.4	310.1
MSDP – M	11.54	660.8	311.4
[%]	18.57	−12.06	0.42
MSDP – A	11.33	677.1	331.0
[%]	16.45	−9.89	6.75

Table 6
Simulation results – water releases and efficiency for Serra da Mesa HP.

	Discharge (m^3/s)	Spillage (m^3/s)	Efficiency (%)	Productivity ($MW/m^3/s$)
DP – PF	774.4	0.0	100.00	1.9710
MSDP – M	741.1	35.6	95.70	1.8579
[%]	−4.30	–		−5.74
MSDP – A	744.6	31.8	96.15	1.8730
[%]	−3.85	–		−4.97

Table 7
Simulation results – operation cost and hydro generation for Serra da Mesa HP plant.

	Cost ($)	Hydro generation (MW)	
		Average	Standard deviation
DP – PF	2.25	832.2	155.1
MSDP – M	2.64	790.5	151.8
[%]	17.11	−5.01	−2.14
MSDP – A	2.61	797.5	165.3
[%]	15.97	−4.16	6.57

Table 9
Simulation results – operation cost and hydro generation for Sobradinho HP.

	Cost ($)	Hydro generation (MW)	
		Average	Standard deviation
DP – PF	2.14	612.5	133.7
MSDP – M	2.48	583.2	156.9
[%]	15.89	−4.79	17.28
MSDP – A	2.47	583.6	154.8
[%]	15.41	−4.73	15.77

Tables 2 and 3 summarizes the simulation results in terms of average values and percentage ratios of water releases, productivity, generation and operating costs, for Furnas hydro plant.

The efficiency is the percentage ratio between spillage and discharge, meanwhile the productivity is the generation, in MW, divided by the inflow, in m^3/s. Percentage values, in bold, are relative to dynamic programming with perfect forecast (DP-PF) which is the reference solution for evaluating the performance of both standard and proposed approaches.

Comparing only the appropriate use of water, MSDP-A presents significant improvements, compared to MSDP-M. The spillage, 2.6 m^3/s lower than in MSDP-M, in average, gives MSDP-A an advantage of 6500 h m^3 of water during the simulation period that represents approximately 34% of the total storage capacity of the reservoir. This smaller spillage and higher storage have guaranteed a better productivity.

Moreover, the reduction in average costs is about 1.5%, compared to MSDP-M which represents, according to an average prices of US$ 52.85/MW h for electrical energy, about US$ 4.7 million per year, totalizing more than US$ 380 million over the planning period. The next two tables show the same simulation results for Foz do Areia hydro plant (Tables 4 and 5).

Water releases in Foz do Areia hydro plant are also better distributed with MSDP-A when compared to MSDP-M. In terms of spillage, MSDP-A average results are 8 m^3/s smaller than MSDP-M what represents, approximately, 523% of the reservoir storage accumulated over the whole planning horizon. The MSDP-A's average reservoir storage is also higher than MSDP-M's, resulting in a 2.12% cost reduction. Tables 6–9 show the results for Serra da Mesa and Sobradinho hydro plants.

The overall results observed for Furnas and Foz do Areia hydro plants are also observed for Serra da Mesa and Sobradinho hydro plants. MSDP-A provides lower spillages than MSDP-M, with higher

Table 8
Simulation results – water releases and efficiency for Sobradinho HP.

	Discharge (m^3/s)	Spillage (m^3/s)	Efficiency (%)	Productivity ($MW/m^3/s$)
DP – PF	2555.6	99.5	96.25	0.3264
MSDP – M	2419.7	237.1	91.13	0.3061
[%]	−5.32	138.25		−6.23
MSDP – A	2476.2	181.7	93.26	0.3137
[%]	−3.11	82.54		−3.91

average reservoir storage and lower costs. The differences, in operation costs, between MSDP-A and MSDP-M, are 1.14% for Serra da Mesa hydro plant and 0.4% for Sobradinho hydro plant.

4. Conclusion

This paper proposes the use of aggregated inflows on stochastic dynamic programming (SDP) for long-term hydrothermal scheduling (LTHS). Aggregated inflows in different time scale resolutions are previously modeled as a Markovian chain into the SDP process (MSDP). The proposed approach, MSDP-A, then uses a nested composition of MSDP models in different time scales up to one year ahead and an inner deterministic optimization step.

The proposed approach, MSDP-A, was compared with two different techniques: MSDP-M, the standard method on monthly basis and PF, the optimal deterministic solution, assuming perfect inflow information. Simulation case studies were conducted for single reservoir hydrothermal systems based on four real Brazilian hydro plants, located in different geographic regions and under distinct hydrological regimes.

Behavior similarities between the two techniques on reservoir storage, discharge, spillage, hydro generation, productivity and operation costs were found. However, the average values of reservoir storage and generation in MSDP-A are higher than in MSDP-M for all simulated hydrothermal systems, which leads to lower costs. Best result among the presented case studies, Foz do Areia hydro plant, has surpassed 2.1% in cost reduction with MSDP-A when compared to MSDP-M.

References

[1] H. Fahlbusch, Early dams, Proceedings of ICE – Engineering History and Heritage 162 (2009) 13–18.
[2] R.E. Bellman, Dynamic Programming, vol. I, Princeton University Press, Princeton, 1957.
[3] D.P. Bertsekas, Dynamic Programming Stochastic Control, Academic Press, New York, 1976.
[4] J.R. Stedinger, B.F. Sule, D.P. Loucks, Stochastic dynamic programming models for reservoir operation optimization, Water Resour. Res. 20 (11) (1984) 1499–1505.
[5] P.J. Werbos, Approximate dynamic programming for real-time control and neural modeling, in: D.A. White, D.A. Sofge (Eds.), Handbook of Intelligent Control, Multiscience Press, New York, 1992, pp. 493–525.
[6] D.P. Bertsekas, J.N. Tsitsiklis, Neuro-Dynamic Programming, Athena Scientific, Belmont, MA, USA, 1996.
[7] A. Al-Tamimi, F.L. Lewis, M. Abu-Khalaf, Discrete-time nonlinear HJB solution using approximate dynamic programming: convergence proof, IEEE Trans. Syst. Man Cybern. B 38 (August (4)) (2008) 943–949.
[8] T. Dierks, B.T. Thumati, S. Jagannathan, Optimal control of unknown affine nonlinear discrete-time systems using offline-trained neural networks with proof of convergence, Neural Netw. 22 (August (5–6)) (2009) 851–860.
[9] R. Enns, J. Si, Helicopter trimming and tracking control using direct neural dynamic programming, IEEE Trans. Neural Netw. 14 (July (4)) (2003) 929–939.
[10] G.K. Venayagamoorthy, R.G. Harley, D.C. Wunsch, Comparison of heuristic dynamic programming and dual heuristic programming adaptive critics for neurocontrol of a turbogenerator, IEEE Trans. Neural Netw. 13 (May (3)) (2002) 764–773.
[11] J. Fu, H. He, X. Zhou, Adaptive learning and control for MIMO system based on adaptive dynamic programming, IEEE Trans. Neural Netw. 22 (July (7)) (2011) 1133–1148.
[12] W.B. Powell, Approximate Dynamic Programming: Solving the Curses of Dimensionality (Probability and Statistics), 2nd ed., Wiley, Hoboken, NJ, USA, 2011.
[13] H. Zhang, D. Liu, Y. Luo, D. Wang, Adaptive Dynamic Programming for Control—Algorithms and Stability, Springer-Verlag, London, U.K, 2013.
[14] Q. Wei, D. Liu, Stable iterative adaptive dynamic programming algorithm with approximation errors for discrete-time nonlinear systems, Neural Comput. Appl. 24 (May (6)) (2014) 1355–1367.
[15] Q. Wei, D. Wang, D. Zhang, Dual iterative adaptive dynamic programming for a class of discrete-time nonlinear systems with time delays, Neural Comput. Appl. 23 (December (7–8)) (2013) 1851–1863.
[16] J.W. Labadie, Optimal operation of multireservoir systems: state of the art review, J. Water Resour. Plan. Manage. 130 (March (2)) (2004) 93–111.
[17] Y. Kim, H. Eum, E. Lee, I. Ko, Optimizing operational policies of a Korean multireservoir system using sampling stochastic dynamic programming with ensemble streamflow prediction, Water Resour. Plan. Manage. 133 (January (1)) (2007) 4–14.
[18] M.S. Zambelli, I. Luna, S. Soares, Long-term hydropower scheduling based on deterministic nonlinear optimization and annual inflow forecasting models, in: Proc. 2009 IEEE Power Tech Conf., Bucharest, 2009, pp. 1–8.
[19] R.O.C. Scarcelli, M.S. Zambelli, S. Soares, A.A.F.M. Carneiro, Aggregated inflows on stochastic dynamic programming for long term hydropower scheduling, in: North American Power Symposium (NAPS), Pullman, 2014, pp. 1–6.
[20] J.A. Tejada-Guibert, S.A. Johnson, J.R. Stedinger, The value of hydrologic information in stochastic dynamic programming models of a multireservoir system, Water Resour. Res. 31 (October (10)) (1995) 2571–2579.
[21] T.A. McMahon, R.M. Vogel, M.C. Peel, G.G.S. Pegram, Global streamflows – part 1: characteristics of annual stream-flows, J. Hydrol. 347 (2007) 243–259.
[22] T.A. McMahon, R.M. Vogel, G.G.S. Pegram, M.C. Peel, D. Etkin, Global streamflows, part 2: reservoir storage—yield performance, J. Hydrol. 347 (2007) 260–271.
[23] T.A. McMahon, M.C. Peel, R.M. Vogel, G.G.S. Pegram, Global streamflows – part 3: country and climate zone characteristics, J. Hydrol. 347 (2007) 272–291.
[24] E. Sutlović, S. Čujić Čoko, I. Medic, Characteristics of basin inflows—a statistical analysis for long-term/mid-term hydrothermal scheduling, in: Proceedings of SDEWES 2013, Dubrovnik, Croatia, September, 2013.
[25] T.G. Siqueira, M.S. Zambelli, M.A. Cicogna, M. Andrade, S. Soares, Stochastic dynamic programming for long term hydrothermal scheduling considering different streamflow models, PMAPS 2006 – 9th International Conference on Probabilistic Methods Applied to Power Systems (2006).
[26] J. Nocedal, S. Wright, Numerical Optimization, 2nd ed., Springer, New York, 2006.